

DIPLOMATIC WORKING DOCUMENT — MODULAR  
ANNEX

ANNEX YA

# Irreversible Disposition Module

*IDM — Surgical Bridge Annex for Irreversible Transformation on Iranian Soil*

MODULAR · INCORPORABLE BY REFERENCE · STEP-BACK / EARN-  
BACK INTEGRATED

Version 1.0 — June 2026

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## **DIPLOMATIC WORKING DOCUMENT — MODULAR ANNEX**

### **Annex Ya**

#### **Irreversible Disposition Module**

IDM — Surgical Bridge Annex for Irreversible Transformation on Iranian Soil

Modular • Incorporable by Reference • Step-Back / Earn-Back Integrated

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### **Preamble**

#### **Purpose**

This Annex Ya creates a single, surgical bridge mechanism — the Irreversible Disposition Module ("IDM") — that resolves the final irreversibility impasse in the MOU Pertaining to PEACE. The IDM performs one irreversible transformation step on Iran's enriched uranium stockpile inside Iran, under full Iranian sovereignty, operated solely by the IAEA, with dual-key activation requiring both the IAEA and a U.S.-designated technical agent. The material never leaves Iran at any stage of the process.

The IDM is the smallest possible political compromise that simultaneously satisfies Iran's fatwa, Iran's sovereignty requirements, the U.S. requirement for a non-Iranian-controlled irreversible step, IAEA verification authority, and the Swiss-Omani escrow symmetry architecture established in Annexes Y and Z.

#### **Background**

The parent framework — the Memorandum of Understanding Pertaining to PEACE (v7.0) — establishes in Article 3.2 that final disposition of enriched uranium stockpiles requires a certified irreversible transformation step before Annex Z pathways (N, O, T, or W) activate. The framework architecture requires that this irreversible step occur under conditions that neither Party controls unilaterally and that the IAEA certifies as a genuine point of no return.

Preceding negotiations produced agreement on the principle of an irreversible step but not on its mechanism. The U.S. requires that the irreversible step not be subject to Iranian unilateral reversal. Iran requires that material remain on Iranian soil under Iranian sovereignty and that no U.S. custody arises at any stage. The IAEA requires full operational

control of the transformation process and clear verification authority. This Annex provides the mechanism that satisfies all three requirements simultaneously.

## **Relationship to Parent Framework**

This Annex Ya operates between Annex Y (Dual-Key Custody Protocol) and Annex Z (In-Iran Transformation Pathways). Material exits the Annex Y Dual-Key Vault, passes through the IDM for the irreversible step under this Annex Ya, and then enters any Annex Z pathway at Iran's unilateral discretion. This Annex does not replace or amend Annex Y or Annex Z; it bridges the operational gap between them.

## **The Immutable Present-Tense Matrix (v2.5)**

This Annex Ya operates under the Immutable Present-Tense Matrix v2.5. All clause construction adheres to the following governing constraints: The framework architecture excludes the vocabulary items "shall," "must," "can," "may," "should," and "will" from all operational provisions. Every sentence carries a clear, active subject performing a direct, present-tense action. The framework architecture excludes passive constructions, hierarchical ordering phrases, and unilateral restraint language. All automated consequences activate as immediate, real-time operational realities. Every prohibitive commitment operates as a feature of the shared framework architecture, not as a concession or voluntary limitation by any single Party.

## **Modularity Clause**

This Annex Ya is self-contained and incorporates into the Memorandum of Understanding Pertaining to PEACE and any successor framework by a single sentence of adoption. It does not amend or supersede conflicting provisions of any parent framework unless the Parties explicitly so agree. In the event of inconsistency, this Annex Ya governs with respect to the irreversible transformation step, IDM facility operations, dual-key IDM activation, IAEA certification of the point of no return, escrow trigger linkage, and the boundary between Annex Y custody and Annex Z pathway activation.

## **OPERATIVE AND BINDING**

# **Article 1: Location and Sovereignty**

## **1.1 IDM Site Designation**

The Irreversible Disposition Module operates inside the Islamic Republic of Iran. Iran and the IAEA jointly designate the IDM site. The designation process requires written agreement between the AEOI Director General and the IAEA Director General. The Parties complete site designation within 30 days of this Annex entering into force.

Iran proposes at least two candidate sites, each meeting the following minimum criteria:

- Structural integrity sufficient to house the IDM equipment as specified in Annex Ya-1 (Technical Equipment Specifications).
- Electrical power supply rated to IDM operational requirements (minimum 500 kW uninterrupted).
- Secure perimeter amenable to IAEA continuous monitoring infrastructure installation.
- Proximity to the Annex Y Dual-Key Vault at Esfahan or mutually agreed alternative, to minimize material transport distance inside Iran.
- Road or rail access for IAEA equipment delivery under AEOI-escorted logistics.

The IAEA evaluates candidate sites against these criteria and confirms technical suitability within 14 days of Iran's submission. If neither candidate site qualifies, the Parties nominate additional sites through the PCC, which convenes within 48 hours.

## **1.2 Iranian Sovereignty**

Iran retains full, undiminished sovereignty over the IDM site and all surrounding territory. This sovereignty includes:

- Exclusive authority over all personnel entering the IDM site, subject only to the access rules in Article 2 of this Annex.
- Control of the outer security perimeter and all access corridors to the IDM facility.
- The right to suspend IDM operations for up to 72 hours in response to verified security threats, with automatic notification to the IAEA and all Parties through the Shared Digital Registry within 15 minutes of the suspension decision.
- Jurisdiction over any criminal or civil matters arising outside the IDM operational perimeter.
- The right to inspect all IAEA and contractor equipment entering the site through the joint logistics inspection procedure in Article 2.4.

Iranian sovereignty does not include the right to instruct the IAEA on the technical conduct of the irreversible transformation step, to override the dual-key activation requirement, or

to permit the material to be removed from the IDM operational corridor without completing the irreversible step.

1.3 Permanence of Site Designation

Once the Parties confirm the IDM site, the designation remains in effect for the full operational period of this Annex, which covers all tranches of material processed through the IDM. The Parties modify the site designation only by written mutual agreement at the level of AEOI Director General and IAEA Director General, with 60 days' advance notice to the PCC Secretariat (Geneva).

Article 2: IAEA Operation and Personnel Access

OPERATIVE AND BINDING

2.1 Sole Operator

The IAEA operates the IDM. No other entity — including the United States, any U.S. contractor, any third-party state, or any Iranian government body — exercises operational control over the IDM or its transformation process. The IAEA Director General designates a Senior IDM Inspector as the on-site operational lead for all IDM transformation activities.

The IAEA Senior IDM Inspector holds the following exclusive operational authorities:

- Authorization of each transformation batch prior to activation.
- Continuous monitoring of the irreversible transformation process.
- Certification of the point of no return for each batch.
- Immediate halt authority if the IAEA detects an anomaly, safety risk, or verification gap.
- Issuance of the Irreversibility Certificate that triggers escrow release under Article 8.

2.2 U.S. Personnel Exclusion

No U.S. personnel, contractors, sub-contractors, or observers are present at the IDM site unless Iran provides explicit, time-limited written consent on a per-visit basis. The consent mechanism operates as follows:

| Parameter         | Specification                                      |
|-------------------|----------------------------------------------------|
| Consent Authority | AEOI Director General or designated deputy         |
| Request Lead Time | Minimum 7 calendar days before proposed visit date |
| Consent Duration  | Single visit, maximum 12 hours; specific zone only |

|                      |                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Personnel Cap        | Maximum 5 U.S. non-uniformed civilian technical observers per visit                                                                                                             |
| Permitted Activities | Observation of IAEA procedures only; no direct equipment contact; no material handling; no independent sampling                                                                 |
| Documentation        | Full name, passport number, professional biography submitted 7 days in advance; Iran approves or denies within 72 hours; silence beyond 72 hours constitutes automatic approval |
| Denial Rights        | Iran denies any individual on security grounds with written justification; denial does not constitute a breach and does not trigger step-back                                   |

The default state is U.S. personnel absence. The IDM operates entirely without U.S. presence on the overwhelming majority of operational days. The consent mechanism is a narrow exception, not a routine feature.

### 2.3 IAEA Personnel Staffing

The IAEA assigns a dedicated IDM operational team, physically present at the IDM site throughout all transformation activities. The minimum staffing for an active transformation shift is:

| Role                     | Minimum Count | Responsibilities                                                     |
|--------------------------|---------------|----------------------------------------------------------------------|
| Senior IDM Inspector     | 1             | Operational lead; holds IAEA Key; certifies point of no return       |
| Verification Inspector   | 2             | Continuous monitoring; mass balance; sampling and assay              |
| Equipment Technician     | 1             | IDM equipment operation and maintenance                              |
| Digital Registry Officer | 1             | Real-time Shared Digital Registry logging; encrypted feed monitoring |

The IAEA provides Iran with the full roster of designated IDM personnel, including professional biographies, 21 days before IDM operations commence. Roster updates require 14 days' advance notice to AEOI. Iran approves or objects to any individual within 10 days; silence constitutes approval.

### 2.4 Joint Logistics Inspection

All equipment, materials, and supplies entering the IDM site pass through a joint logistics inspection at a designated checkpoint 1 kilometer from the IDM facility perimeter. The



inspection team consists of one AEOI inspector and one IAEA verification inspector. Both must sign the equipment manifest before any item enters the perimeter.

The IAEA submits the equipment manifest to AEOI at least 72 hours before each scheduled delivery. AEOI may request technical clarification within 48 hours. Any equipment not listed on the approved manifest does not enter the IDM site until a supplementary manifest is approved through the same process.

## Article 3: Dual-Key Activation Architecture

### OPERATIVE AND BINDING

#### 3.1 The Two Keys

IDM activation requires two independent digital keys presented simultaneously within a 90-second authentication window:

| Key                 | Holder                                                                                   | Authorization Scope                                                                                                                                                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IAEA Key            | IAEA Director General or designated Senior IDM Inspector on-site                         | Certifies that radiological safety, verification protocols, and IAEA operational standards are met for the activation batch. Without this key, the IDM does not activate.                                                                             |
| U.S.-Designated Key | Non-uniformed technical agent of the United States, approved by Iran per MOU Article 3.5 | Provides the second factor of irreversibility control. The U.S.-Designated Key holder has no operational authority over the IDM, no material handling rights, and no presence on the IDM site unless Iran grants per-visit consent under Article 2.2. |

#### 3.2 Key Generation and Custody Infrastructure

Both IDM keys generate physically inside a Hardware Security Module (HSM) hosted by the Central Bank of Oman in Muscat. The HSM architecture is identical to the architecture established for the Annex Y Dual-Key Vault. Neither key exists in extractable form outside the HSM. The Central Bank of Oman maintains the physical infrastructure but holds no logical access to either key material.

Key custody arrangements:

- IAEA Key: Primary custody with the IAEA Senior IDM Inspector on-site. Backup key in a geographically isolated secure facility in Vienna.
- U.S.-Designated Key: Primary custody with the designated technical agent. Backup key in Zurich.

- Both Parties audit the HSM configuration and key custody arrangements prior to IDM first activation.
- Key rotation occurs every 90 days or immediately upon a key holder change, whichever comes first. The Shared Digital Registry logs every rotation event.

### 3.3 Remote Activation Protocol

The U.S.-Designated Key activates remotely by default. The U.S. technical agent presents the key from a secure remote terminal at a location agreed with Iran and the IAEA, without physical presence at the IDM site. Remote activation reduces the operational burden of per-visit consent and maintains the security of the dual-key architecture.

Remote activation operates through the following sequence:

1. The IAEA Senior IDM Inspector verifies the transformation batch is ready and submits an activation request to the Shared Digital Registry, logged with an automated timestamp.
2. The Shared Digital Registry notifies the U.S. technical agent's secure terminal within 5 minutes.
3. The U.S. technical agent reviews the batch verification summary — including IAEA mass balance, safety clearance, and chain-of-custody confirmation — through the secure terminal.
4. If the batch summary confirms compliance, the U.S. technical agent presents the U.S.-Designated Key within the 90-second authentication window.
5. The IAEA Senior IDM Inspector presents the IAEA Key within the same 90-second window.
6. The HSM validates both keys simultaneously and generates an activation authorization token, logged in the Shared Digital Registry with a timestamp accurate to the second.
7. The IDM activates. The irreversible transformation step commences.

If the U.S. technical agent does not present the key within 6 hours of the Digital Registry notification for reasons other than a verified technical failure, the IAEA Senior IDM Inspector escalates to the PCC Secretariat (Geneva), which convenes within 24 hours to resolve the delay.

### 3.4 Key Withholding and Dispute Resolution

Neither key holder withholds the key arbitrarily after the IAEA confirms batch compliance. The U.S. technical agent withholds the key only upon documented evidence of one of the

following:

- A discrepancy in the IAEA batch verification summary that the U.S. technical agent flags in writing to the IAEA within 2 hours of notification.
- An active Tier 3 Deliberate Hostile Violation under Annex T that the PCC has not yet resolved.
- A verified technical failure in the IDM equipment that the IAEA Senior IDM Inspector confirms in writing.

Any withholding triggers automatic PCC notification within 15 minutes. The PCC convenes a technical review panel within 48 hours. The review panel issues a determination within 72 hours. A determination of improper withholding by either key holder constitutes a Tier 2 Material Disruption under Annex T.

### 3.5 Single-Key Attempt — Automatic Safeguard

Any attempt to activate the IDM using only one key triggers:

- Immediate IDM lock for 72 hours.
- Automatic Tier 2 Material Disruption classification under Annex T.
- Notification to both Parties, the IAEA, and all guarantor states within 5 minutes through the Shared Digital Registry.
- Logging of the attempt with timestamp, originating key identifier, and attempted activation parameters.

A single-key activation attempt does not itself constitute a Tier 3 violation unless accompanied by evidence of intent to divert material, in which case Tier 3 classification applies immediately.

## Article 4: Material Flow and Chain of Custody

### OPERATIVE AND BINDING

#### 4.1 Entry into the IDM

Material enters the IDM exclusively from the Annex Y Dual-Key Vault. The material flow pathway is:

| Step | Action | Verification |
|------|--------|--------------|
|------|--------|--------------|

|   |                                                                                                                                                                                                        |                                                                                                                                                     |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Iran notifies PCC and IAEA of intended withdrawal from Annex Y Vault for IDM processing, specifying batch size and target transformation method.                                                       | PCC logs notification; IAEA verifies batch parameters are consistent with approved transformation plan.                                             |
| 2 | Both Annex Y Dual-Key holders open the Vault per the Article 3.2 withdrawal procedure of Annex Y.                                                                                                      | Shared Digital Registry logs vault opening with timestamp and batch manifest.                                                                       |
| 3 | IAEA seals each withdrawn container with a unique tamper-evident, cryptographically signed seal. The IAEA assigns a GPS-tracked transport module to each container.                                    | IAEA Vault Withdrawal Certificate issued; seal serial numbers logged in Shared Digital Registry.                                                    |
| 4 | Containers move under continuous IAEA chain-of-custody from the Annex Y Vault to the IDM facility. The transport route stays within Iranian territory. AEOI security escorts the transport throughout. | GPS tracking feeds to IAEA Vienna and the Shared Digital Registry in real time. Any position anomaly triggers automatic PCC alert within 2 minutes. |
| 5 | IAEA Senior IDM Inspector verifies seal integrity on each container at the IDM entry point before any container enters the IDM operational corridor.                                                   | Seal verification logged in Shared Digital Registry; any compromised seal triggers immediate Tier 2 classification.                                 |

## 4.2 Material Stays in Iran

At no stage of the IDM process does material leave Iranian territory. This is an absolute architectural constraint, not a preference. The IDM facility is inside Iran. The Annex Y Vault is inside Iran. All Annex Z transformation pathways (N, O, T, and W) operate inside Iran. The chain-of-custody corridor from vault entry to IDM processing to Annex Z pathway activation runs entirely within Iranian sovereign territory.

Any instruction, proposal, or operational plan that would require material to exit Iranian territory at any stage of this Annex Ya process is inconsistent with this Annex and does not proceed without formal amendment through the PCC and AEOI Director General co-signature.

## 4.3 Batch Sizing and Tranche Architecture

Material processes through the IDM in tranches aligned with the escrow release schedule established in Article 8. The default tranche architecture is:

| Tranche | Material Quantity            | Approximate % of Stockpile | Escrow Trigger                                |
|---------|------------------------------|----------------------------|-----------------------------------------------|
| T-1     | Initial 50 kg HEU-equivalent | ~16%                       | Day-Zero Cryptographic Escrow first release   |
| T-2     | Next 100 kg HEU-equivalent   | ~33%                       | Second tranche escrow release per Article 8.2 |

|     |                                           |      |                                                 |
|-----|-------------------------------------------|------|-------------------------------------------------|
| T-3 | Next 100 kg HEU-equivalent                | ~33% | Third tranche escrow release per Article 8.2    |
| T-4 | Remaining balance (approximately 60.9 kg) | ~20% | Final completion escrow release per Article 8.3 |

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The total nominal stockpile subject to this Annex Ya is 310.9 kg of 60%-enriched HEU held in the Annex Y Dual-Key Vault, consistent with the baseline established in MOU Article 3.1. The Parties adjust tranche quantities by mutual written agreement through the PCC if the verified stockpile baseline differs from this nominal figure after IAEA mass spectrometry confirmation.

### 4.4 Continuous Chain-of-Custody Logging

The Shared Digital Registry (Annex U) maintains a continuous, timestamped, tamper-evident log of every material movement event from Annex Y Vault withdrawal to IDM entry to post-IDM transfer to Annex Z pathway. Every log entry includes: container identifier, seal serial number, GPS coordinates, IAEA officer signature, timestamp accurate to the second, and the active key authorization token.

Live encrypted feeds from the IDM facility transmit to monitoring stations in Tehran, Vienna (IAEA), the PCC Secretariat (Geneva), and the Pakistan/Oman facilitation team nodes simultaneously. No feed interruption exceeds 10 minutes without automatic PCC escalation.

## Article 5: The Irreversible Step

### OPERATIVE AND BINDING

### 5.1 Transformation Methods

The IDM performs one of the following three irreversible transformations per batch, selected by Iran before the tranche is withdrawn from the Annex Y Vault. Iran notifies the PCC and IAEA of the selected method at the time of vault withdrawal notification under Article 4.1 Step 1. Iran selects the method for each tranche independently; different tranches proceed through different methods.

| Method | Technical Description | Point of No Return |
|--------|-----------------------|--------------------|
|--------|-----------------------|--------------------|

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|                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Method A —<br>Downblending to<br>Natural Uranium<br>Equivalent ( $\leq 0.7\%$<br>U-235) | The IDM dilutes the 60% HEU with depleted uranium hexafluoride (UF6) or natural uranium tetrafluoride (UF4) feed material in a precision-controlled blending cascade. The cascade operates at sub-critical geometry. IAEA continuous gamma spectroscopy confirms that the U-235 isotopic fraction of the product stream drops below and remains below 0.7% throughout the blending cycle. | The point of no return is the moment the product stream first verifies below 0.7% U-235 by IAEA mass spectrometry, which the IAEA records as the Certification Timestamp. Reconstitution to weapons-usable enrichment levels would require dedicated enrichment infrastructure beyond the scope of any plausible clandestine program. |
| Method B —<br>Chemical<br>Immobilization                                                | The IDM converts the HEU to a stable, chemically immobilized form — uranium oxide (U3O8) ceramic pellets or borosilicate glass matrix — through a controlled calcination or vitrification process. The product is physically and chemically stable, environmentally contained, and suitable for long-term storage or direct Pathway T/W processing.                                       | The point of no return is the completion of calcination or vitrification, at which stage reconstitution to metal or UF6 form requires multi-stage chemical processing detectable by the IAEA with near-100% probability under Complementary Access protocols.                                                                         |
| Method C —<br>Metallurgical<br>Destruction                                              | The IDM alloys the HEU with zirconium or stainless steel in a controlled metallurgical furnace, producing a dilute uranium alloy with U-235 concentration far below weapons-grade. The alloy is cast into defined geometric forms and inventoried by IAEA serial number.                                                                                                                  | The point of no return is the completion of alloying and casting, after which separation of the uranium from the alloy matrix requires chemical dissolution processing that the IAEA detects immediately through its environmental monitoring suite.                                                                                  |

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## 5.2 IAEA Certification of the Point of No Return

After each batch completes the irreversible transformation step, the IAEA Senior IDM Inspector:

8. Performs mass spectrometry and isotopic assay on representative samples from the transformed product.
9. Compares the assay results against the pre-transformation baseline established in the batch verification summary.
10. Confirms that the transformation method achieved the specified outcome (U-235 concentration, chemical form, or alloy composition as applicable).
11. Issues a Point-of-No-Return Certificate (PONR Certificate) for the batch, recording: batch identifier, transformation method, pre-transformation mass and enrichment, post-transformation mass and enrichment, IAEA officer name and credential, and Certification Timestamp.

12. Logs the PONR Certificate in the Shared Digital Registry within 2 hours of the Certification Timestamp.

The PONR Certificate is the sole instrument that triggers the corresponding escrow release under Article 8. No political body — including the PCC, the U.S. Treasury, or the AEOI — overrides, delays, or conditions a duly issued PONR Certificate.

### 5.3 Transformation Quality Assurance

The IAEA conducts the following quality assurance procedures for each transformation batch:

- Pre-transformation baseline assay: mass spectrometry confirming 60% HEU enrichment before the IDM activation key sequence.
- Mid-process sampling: for Method A, continuous gamma spectroscopy readings every 15 minutes during the blending cascade. For Methods B and C, temperature and chemical composition monitoring at 30-minute intervals.
- Post-transformation assay: mass spectrometry and independent radiological assay within 4 hours of transformation completion.
- Mass balance reconciliation: IAEA confirms that post-transformation product mass (adjusted for process losses within certified tolerance) accounts for 100% of pre-transformation input mass.

Mass balance tolerance is  $\pm 0.3\%$  for Methods A and B, and  $\pm 0.5\%$  for Method C (accounting for metallurgical processing losses). Any discrepancy beyond tolerance triggers Tier 2 classification under Annex T and halts further IDM activations pending PCC technical review.

### 5.4 No Reversibility Permitted

The framework architecture excludes any transformation process that the IAEA classifies as reversible in its Technical Assessment Protocol for IDM methods. The IAEA's classification is final and not subject to political override. If Iran proposes a transformation method variant that the IAEA classifies as reversible, the IDM does not activate with that variant until the IAEA issues a revised technical clearance confirming irreversibility under the modified parameters.

## Article 6: U.S. Non-Custody Architecture

### OPERATIVE AND BINDING

## **6.1 No U.S. Physical Custody**

At no point in the IDM process does the United States obtain physical custody of any material subject to this Annex Ya. Physical custody is defined as: direct physical possession, handling, or control of a container, sample, or any quantity of the material, whether by U.S. government personnel, U.S. uniformed or non-uniformed military, U.S. contractors, or U.S.-funded third parties acting on U.S. government instruction.

The U.S.-Designated Key does not constitute physical custody. The key is a digital authorization mechanism. Its holder activates a remote signal that enables the IAEA to commence a transformation process that the IAEA alone executes. Activation of the key does not transfer, and the framework architecture excludes any interpretation that would treat key activation as a form of custody.

## **6.2 No U.S. Legal Custody**

At no point in the IDM process does the United States acquire legal title to, legal responsibility for, or legal rights over any material subject to this Annex Ya. The material remains Iranian property throughout. IAEA operational control during the transformation step is a verification mechanism, not a transfer of legal title. The United States acquires no legal rights or responsibilities with respect to the material as a result of any action taken under this Annex Ya.

## **6.3 No U.S. Operational Control**

The U.S.-Designated Key holder exercises no operational control over the IDM, its equipment, its transformation parameters, its sequencing, or its output. The sole operational decisions within the IDM process that the U.S.-Designated Key holder influences are the binary decisions to present or withhold the key. Grounds for withholding are enumerated in Article 3.4. All other operational decisions are the exclusive province of the IAEA.

## **6.4 Affirmative Statement of Non-Custody**

The Parties confirm that the IDM architecture is designed to satisfy, and the Parties jointly represent that it does satisfy, the following conditions:

- Iran's fatwa prohibiting weapons of mass destruction and their associated technologies: The IDM performs no weapons-enabling process. The irreversible transformation step reduces weapons-usable material to sub-weapons-grade form or a chemically



inaccessible form. No component of the IDM architecture aids, assists, or advances weapons development.

- Iran's sovereignty: The material stays in Iran. IAEA operational control of the transformation step is time-limited, technically specific, and does not constitute occupation, servitude, or encumbrance on Iranian sovereignty beyond the specific operational perimeter of the IDM facility during active transformation.
- U.S. requirement for non-Iranian-controlled irreversibility: The irreversible step requires the IAEA key. The IAEA is independent of Iran. Neither Iran alone nor the U.S. alone controls the transformation process. The dual-key architecture and IAEA operational authority constitute the non-Iranian control the U.S. requires.
- No U.S. custody at any stage: The architecture excludes U.S. physical custody, legal custody, and operational control. The U.S.-Designated Key is a verification and confirmation mechanism. No U.S. actor touches, holds, or legally acquires any material at any stage.

# Article 7: Relationship to Annex Z Pathways

## OPERATIVE AND BINDING

### 7.1 IDM as Gateway

The IDM is the gateway between the Annex Y custody phase and the Annex Z production phase. Material that completes the IDM irreversible transformation step proceeds to any Annex Z pathway at Iran's unilateral discretion. No U.S. concurrence, PCC vote, or additional authorization is required for Iran's Annex Z pathway selection after IDM processing is complete.

### 7.2 Pathway Compatibility

All four Annex Z pathways are compatible with all three IDM transformation methods:

| IDM Method                       | Pathway N<br>Compatibility                                                                 | Pathway O<br>Compatibility                                                                   | Pathway T / W<br>Compatibility                                                 |
|----------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Method A — Downblending to ≤0.7% | Direct: LEU suitable for further downblending to reactor-grade in Annex Z Pathway N cycle. | Direct: ≤0.7% feed material is the standard input for Pathway O industrial uranium products. | Compatible via additional processing step in the Pathway T/W production chain. |

|                                            |                                                                                                                                                               |                                                                                                   |                                                                                                    |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Method B —<br>Chemical<br>Immobilization   | Requires chemical reconversion to UF6 before Pathway N use; IAEA verifies reconversion as non-reversible to weapons-grade.                                    | U3O8 ceramic is the standard form for industrial uranium products; direct pathway.                | Optimal form for Pathway T depleted uranium industrial use and Pathway W agriculture applications. |
| Method C —<br>Metallurgical<br>Destruction | Not directly compatible; alloy material requires acid dissolution before nuclear fuel cycle re-entry, which the IAEA treats as a separate authorized process. | Compatible; uranium-metal alloys are a recognized industrial feedstock category within Pathway O. | Compatible via alloy product applications in Pathway T industrial uses.                            |

Iran selects the Annex Z pathway and notifies the PCC within 7 days of the IAEA issuing the PONR Certificate for each batch. The Annex Z pathway commences according to its own operational timeline in Annex Z once the IDM-processed material is transferred under IAEA chain-of-custody to the designated Annex Z facility.

**7.3 No U.S. Veto Over Annex Z Selection**

Iran's right to select any Annex Z pathway after IDM processing is complete is unilateral and irrevocable, consistent with MOU Article 3.2 and Article 1.9. The United States has no veto, no consultation right beyond technical information, and no ability to delay or block Iran's Annex Z pathway selection. This Annex Ya does not add any condition or qualification to the unilateral Annex Z selection right established in the parent framework.

**7.4 Joint Disposition Fund Access Post-IDM**

Iran's access to the Joint Disposition Fund drawdown schedule under Annex Z (Article 1.4 of Annex Z) activates upon IAEA issuance of the PONR Certificate under this Annex Ya, not upon completion of any Annex Z pathway milestone. The PONR Certificate constitutes the milestone that unlocks Annex Z fund drawdown eligibility for the corresponding tranche. Annex Z then governs the specific drawdown amounts and timeline milestones for each pathway.

**7.5 Pathway W Exemption from IDM Gateway**

Pathway W (Agriculture and Food Security), as defined in Annex Z, operates independently of the Irreversible Disposition Module. Material allocated to Pathway W moves directly from the Annex Y Dual-Key Vault to Pathway W processing under the operational sequencing of Annex Y.b. The IDM dual-key activation requirement does not apply to Pathway W.

The IAEA verifies the irreversible destruction of Pathway W material through isotopic sampling, continuous monitoring, and field application protocols set out in Annex Y.b, Article 5.2. The United States holds no veto or key over Pathway W destruction.

This exemption does not affect the application of the IDM to any other Annex Z pathway. Pathways N, O, and T proceed through the IDM gateway under Articles 7.1 through 7.4 of this Annex. Pathway W bypasses the IDM entirely. The framework architecture treats Pathway W as a separate, parallel irreversibility track that satisfies the parent MOU’s irreversibility requirement through its own dedicated verification architecture.

# Article 8: Escrow Symmetry and Cryptographic Trigger

## OPERATIVE AND BINDING

### 8.1 Clean Escrow Trigger Architecture

Activation of the IDM — meaning the moment both dual keys are presented and the HSM generates the activation authorization token — triggers a conditional hold release flag in the Day-Zero Cryptographic Escrow registry for the corresponding tranche. The release flag activates automatically. No political decision, no further authorization, and no additional procedural step is required to set the flag.

The PONR Certificate issued under Article 5.2 is the second and final trigger. When both conditions are satisfied — IDM activation timestamp and PONR Certificate timestamp — the Swiss-Omani escrow mechanism releases the corresponding tranche according to the release schedule in Article 8.2.

### 8.2 Tranche Release Schedule

| Tranche | Trigger Condition                                        | Release Amount | Release Timeline                                                                                                            |
|---------|----------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| T-1     | IDM activation of first 50 kg batch + PONR Certificate   | \$750 million  | Swiss facilitator releases within 24 hours of PONR Certificate logging in Shared Digital Registry.                          |
| T-2     | IDM activation of second 100 kg batch + PONR Certificate | \$1.5 billion  | Swiss facilitator releases within 24 hours of PONR Certificate. Includes Joint Disposition Fund initial drawdown component. |

|             |                                                          |                                                              |                                                                                                                                                                                  |
|-------------|----------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-3         | IDM activation of third 100 kg batch + PONR Certificate  | \$1.5 billion                                                | Swiss facilitator releases within 24 hours of PONR Certificate.                                                                                                                  |
| T-4 (Final) | IDM activation of final balance batch + PONR Certificate | Remaining escrow balance plus \$500 million completion bonus | Swiss facilitator releases within 48 hours of final PONR Certificate. Completion bonus: 50% to Iran Central Bank; 50% to designated humanitarian escrow administered by UN OCHA. |

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### 8.3 Swiss-Omani Escrow Symmetry

The escrow architecture for this Annex Ya maintains the Swiss-Omani symmetry established in Annex Y:

- The Swiss facilitator administers the escrow account and executes all tranche releases.
- The Central Bank of Oman hosts the escrow account under sovereign immunity protections.
- The Pakistan/Oman facilitation team serves as the independent conduit for escrow status notifications to Iran and the U.S.
- Russia serves as the independent escrow balance auditor, receiving real-time notification of every deposit, release, and balance change.
- No political entity — including the U.S. Treasury, the AEOI, or the PCC — overrides an escrow release triggered by a duly issued PONR Certificate.

This escrow symmetry architecture ensures that neither Party can condition, delay, or redirect a release after the IAEA PONR Certificate triggers the automatic release sequence. The architecture is identical in its fundamental structure to the existing Annex Y escrow mechanism, permitting both sides to rely on the same institutional relationships and legal instruments they already accepted.

### 8.4 No-Clawback Provision

Funds released from escrow upon PONR Certificate trigger are not subject to clawback, rescission, or hold by any Party, for any reason, under any circumstances, including:

- A subsequent Tier breach under Annex T, provided the breach occurs after the PONR Certificate timestamp for the released tranche.
- A change in government, executive order, or legislative action by either Party.
- A dispute about the technical accuracy of any PONR Certificate, provided the dispute arises after the release executes. Pre-release disputes proceed through the Article 3.4 withholding and dispute resolution mechanism.

The no-clawback provision is absolute and non-waivable. It applies identically to all four tranches and to the completion bonus.

8.5 Humanitarian Firewall

The humanitarian component of the completion bonus (50% of the \$500 million completion bonus, administered by UN OCHA) enjoys protected status under the Tier breach architecture of Annex T. No Tier 1, Tier 2, or Tier 3 classification reduces, holds, or redirects humanitarian funds once released.

Article 9: Step-Back / Earn-Back Integration

OPERATIVE AND BINDING

9.1 Governing Framework

This Annex Ya integrates the Three-Tier Breach and Step-Back Architecture established in Annex T. The tier system governs all IDM-specific breach classifications, penalty assessments, and earnback procedures. The framework architecture excludes kinetic enforcement mechanisms. All penalties are financial and diplomatic.

9.2 IDM-Specific Tier Thresholds

| Tier                             | IDM Telemetry Signature                                                                                                                                                                                                                | Consequence                                                                                                                              | Remediation Window |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Tier 1<br>Technical<br>Dispute   | Mass balance discrepancy below 0.3% tolerance; IAEA reporting delays under 48 hours; batch verification delay under 24 hours; minor equipment calibration deviation self-corrected within the shift.                                   | No tranche hold. PCC technical notification. 14-day resolution window. Quiet compliance correction.                                      | 14 days            |
| Tier 2<br>Material<br>Disruption | Mass balance discrepancy above 0.3% tolerance; single-key activation attempt; U.S. key withholding beyond 6 hours without documented grounds; unauthorized equipment in IDM corridor; tamper-evident seal compromise on any container. | Graduated benefit reduction per Annex T Article 4.2. IDM operations pause pending PCC technical review panel. 30-day remediation window. | 30 days            |

|            |                                               |                                |      |
|------------|-----------------------------------------------|--------------------------------|------|
| Tier 3     | Diversion of material from IDM corridor; IAEA | Full escrow freeze for nuclear | 90   |
| Deliberate | Senior IDM Inspector obstruction or removal;  | module. Immediate referral to  | days |
| Hostile    | destruction of monitoring infrastructure with | Guarantor Matrix. UNSC         |      |
| Violation  | apparent intent; unauthorized transport of    | notification within 24 hours.  |      |
|            | material outside Iran; material removed from  | 90-day remediation window.     |      |
|            | IDM without completing irreversible step.     |                                |      |

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### 9.3 Earn-Back Architecture

Full earn-back from any Tier breach classification requires:

- Tier 1: Verified technical correction within the 14-day window and IAEA clean certification. No financial impact; automatic restoration upon clean certification.
- Tier 2: Verified correction of the specific breach condition within the 30-day window, 30 consecutive days of clean IDM telemetry, and IAEA clean certification. Financial reductions reinstate upon PCC confirmation of remediation completion.
- Tier 3: 90 consecutive days of verified clean IDM operations, IAEA clean certification, PCC majority confirmation, and written assurance from the breaching Party's head of government that the breach conditions are fully remediated and non-recurrent.

Humanitarian tranches remain protected throughout any earn-back period. The framework architecture excludes humanitarian funds from reduction, hold, or delay regardless of tier classification.

## Article 10: Relationship to Other Annexes

### OPERATIVE AND BINDING

#### 10.1 Annex Y — Dual-Key Custody Protocol

This Annex Ya activates after Annex Y custody is established and material is ready for transformation processing. Annex Y governs all material during the custody phase prior to IDM withdrawal. Upon vault withdrawal under Article 4.1 of this Annex, this Annex Ya governs. Upon PONR Certificate issuance, governance of the transformed material shifts to Annex Z for the selected pathway. Annex Y continues to govern any material remaining in the vault that has not yet been withdrawn for IDM processing.

#### 10.2 Annex Z — In-Iran Transformation Pathways

This Annex Ya is the prerequisite gateway for Annex Z pathway activation. Material does not enter any Annex Z pathway without first completing the IDM process and receiving a

PONR Certificate under this Annex Ya. After PONR Certificate issuance, Annex Z governs exclusively. This Annex Ya has no further role in the material's disposition after the PONR Certificate timestamp.

### **10.3 Annex T — Step-Back / Earn-Back**

The Three-Tier Breach and Step-Back Architecture of Annex T governs all breach assessments, penalty calculations, and earn-back procedures for obligations arising under this Annex Ya. The IDM-specific tier thresholds in Article 9.2 of this Annex are calibrated sub-specifications within the Annex T framework, not a replacement for it. In any conflict between IDM-specific thresholds in this Annex and general thresholds in Annex T, the IDM-specific thresholds govern with respect to IDM operations.

### **10.4 Annex U — Shared Digital Registry**

The Shared Digital Registry (Annex U) provides the real-time logging, chain-of-custody documentation, key activation records, and PONR Certificate repository for all IDM operations. Every automated trigger, notification, and certification event under this Annex Ya logs in the Shared Digital Registry within the specified timeframes. The Registry is the authoritative record for all dispute resolution proceedings arising from this Annex Ya.

### **10.5 Annex V — Nuclear Dust Protocol**

Where material subject to Annex V (Sub-Surface Remediation and Joint Sovereignty Protocol) is recovered and containerized to IAEA standards, the recovered material proceeds through the IDM under this Annex Ya before entering any Annex Z pathway. The CANDOR chain-of-custody established in Annex V connects to the IDM material flow under Article 4 of this Annex upon transport from the recovery site to the IDM facility. All Annex V sovereignty, personnel exclusion, and escrow provisions remain in effect for the recovery and transport phase; this Annex Ya governs from IDM entry.

## **Article 11: Incorporation by Reference**

This Annex Ya incorporates into the Memorandum of Understanding Pertaining to PEACE (v7.0) and into any successor framework by the following sentence of adoption:

"The Parties adopt Annex Ya (Irreversible Disposition Module, Version 1.0, June 2026) as a binding modular annex to this Agreement, operative from the date of this signature."

No further drafting, amendment, or supplemental agreement is required to give full operational effect to this Annex Ya upon adoption.

Signature Block

| FOR THE UNITED STATES OF AMERICA | FOR THE ISLAMIC REPUBLIC OF IRAN |
|----------------------------------|----------------------------------|
| Signature                        | Signature                        |
| Name and Title                   | Name and Title                   |
| Date                             | Date                             |

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WITNESSED BY

| FOR THE IAEA            | FOR THE PAKISTAN / OMAN FACILITATION TEAM |
|-------------------------|-------------------------------------------|
| Director General — IAEA | Lead Facilitator                          |
| Date                    | Date                                      |

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Annex Ya — Irreversible Disposition Module (IDM)

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End of Annex

Annex Ya — Irreversible Disposition Module (IDM)

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